Data Representation and File Structures

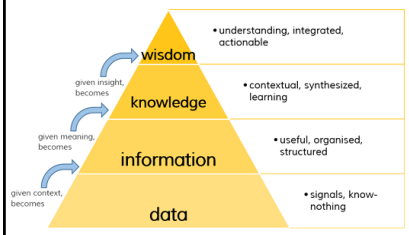
NGUYEN Duc Dung, Ph.D.
Institute of Information Technology, VAST

Informatics: Course Content

- History of Computing and Computer Architecture *(no practice)*
- **Data Representation and File Structures**
- Operating System
- Networks and the Internet
- Software Tools for Techies
- Introduction to Programming and Software Engineering

2

DIKW Pyramid



USTH, 09/12/2019
09/12/2009

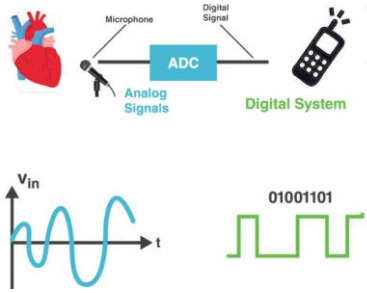
2067/QĐ-TTg
09/12/2009

09/12/2009

09122009

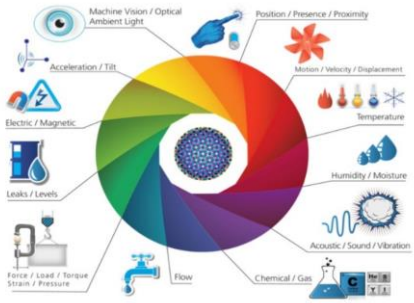
3

Analog to Digital Conversion

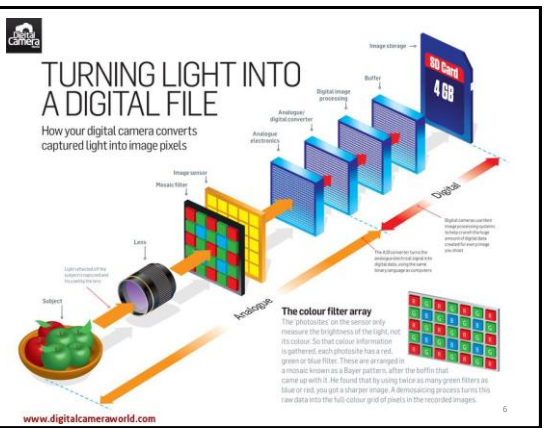


4

Data: from Trillion Sensor World

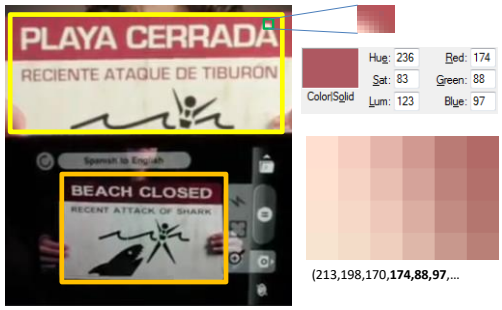


5



6

Image Data



(213,198,170,**174,88,97**,...

7

Text File

Byte Order Mark (BOM):

Encoding	Representation (hexadecimal)	Representation (decimal)	Bytes as CP1252 characters
UTF-8[a]	`EF BB BF`	`239 187 191`	ï»¿
UTF-16 (BE)	`FE FF`	`254 255`	þÿ
UTF-16 (LE)	`FF FE`	`255 254`	ÿþ
UTF-32 (BE)	`00 00 FE FF`	`0 0 254 255`	^@^@þÿ (^@ is the null character)
UTF-32 (LE)	`FF FE 00 00`	`255 254 0 0`	ÿþ^@^@ (^@ is the null character)

8

BMP Image File



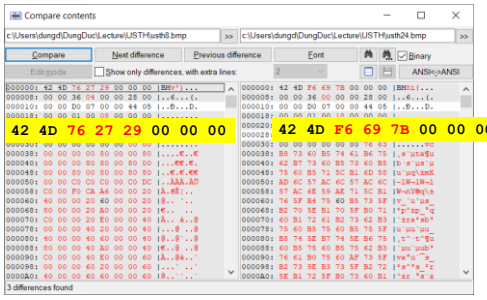
9

BMP: 24-bit vs. 8-bit



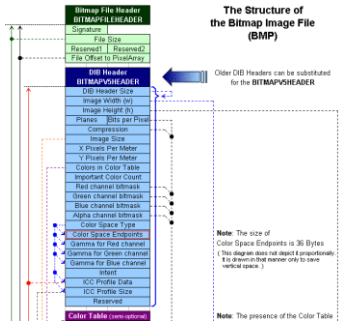
10

BMP: 24-bit vs. 8-bit



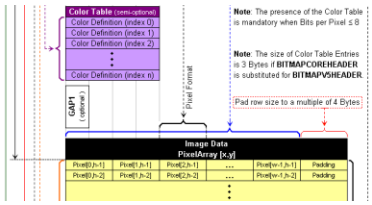
11

BMP Image File Structure



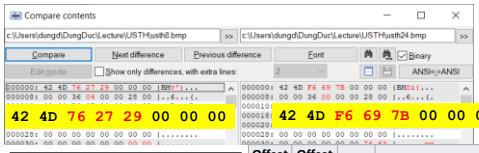
12

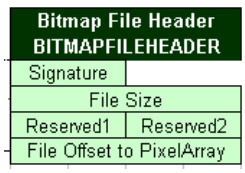
BMP Image File Structure (cont.)



13

BMP: 24-bit vs. 8-bit

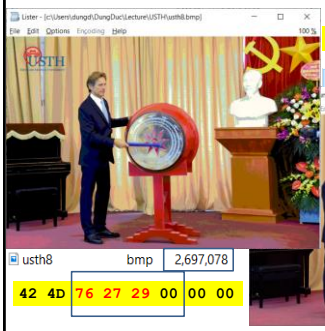




Offset hex	Offset dec	Size	Purpose
00	0	2 bytes	To **identify** the BMP and DIB file is **0x42 0x4D**
02	2	4 bytes	The **size of the BMP file** in bytes
06	6	2 Bytes	Reserved;

14

Data Representation



42 4D F6 69 7B 00 00 00

usth24 bmp 8,088,054

usth8 bmp 2,697,078

42 4D 76 27 29 00 00 00

15

Data Representation

42 4D F6 69 7B 00 00 00

usth24 bmp 8,088,054

usth8 bmp 2,697,078

42 4D 76 27 29 00 00 00

16

Decimal vs. Hexadecimal

Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Dec	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

42 4D F6 69 7B 00 00 00

usth24 bmp 8,088,054

Dec: 2697078
Hex: 00292776

Dec: 8088054
Hex: 007B69F6

usth8 bmp 2,697,078

42 4D 76 27 29 00 00 00

17

Decimal vs. Hexadecimal

Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Dec	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Dec:
2697078 = 8*10^0 + 7*10^1 + 0*10^2 + 7*10^3 + 9*10^4 + 6*10^5 + 2*10^6

42 4D F6 69 7B 00 00 00

usth24 bmp 8,088,054

Dec: 2697078
Hex: 00292776

Dec: 8088054
Hex: 007B69F6

usth8 bmp 2,697,078

42 4D 76 27 29 00 00 00

Hex:

007B69F6 = ?

18

Decimal vs. Hexadecimal

Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Dec	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Dec:
2697078 = 8*10^0 + 7*10^1 + 0*10^2 + 7*10^3 + 9*10^4 + 6*10^5 + 2*10^6

42 4D F6 69 7B 00 00 00

usth24 bmp 8,088,054

Dec: 2697078
Hex: 00292776

Why?

Dec: 8088054
Hex: 007B69F6

Hex:
007B69F6 = 6*10^0 + 15*16^1 + 9*16^2 + 6*16^3 + 11*16^4 + 7*16^5

usth8 bmp 2,697,078

42 4D 76 27 29 00 00 00

19

Decimal vs. Hexadecimal vs. Binary

Why binary?

42 4D F6 69 7B 00 00 00

usth24 bmp 8,088,054

Dec: 2697078
Hex: 00292776
Bin: 10100100100111011110110

Dec: 8088054
Hex: 007B69F6
Bin: 11110110110100111110110

usth8 bmp 2,697,078

42 4D 76 27 29 00 00 00

20

Binary Addition

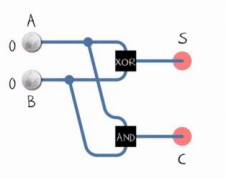
		0 A8	1 A7	0 A6	0 A5	1 A4	1 A3	1 A2	0 A1
	+	0 B8	1 B7	0 B6	0 B5	1 B4	0 B3	1 B2	1 B1
Overflow?	Carried Values	1 C8	C7	C6	1 C5	1 C4	1 C3	C2	
☐	=	1 S8	0 S7	0 S6	1 S5	1 S4	0 S3	0 S2	1 S1

https://www.101computing.net/binary-additions-using-logic-gates/

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Half Adder

Inputs		Outputs	
A	B	C	S
0	0	0	0
1	0	0	1
0	1	0	1
1	1	1	0

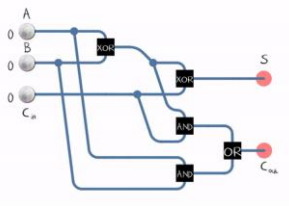


By Marble machine - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=48017339

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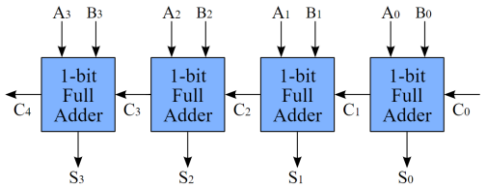
Full Adder

Inputs			Outputs	
A	B	C_{in}	C_{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



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4-bit Adder



Limitation?

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Information and Data

Different Types Data For You



Data For Computer System

```
01110101 00100000
01110010 01101111
01100011 01101011
01111001 01101111
01110101 00100000
01110010 01101111
01100011 01101011
```

https://www.learncomputerscienceonline.com/binary-number-system/

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Information Management

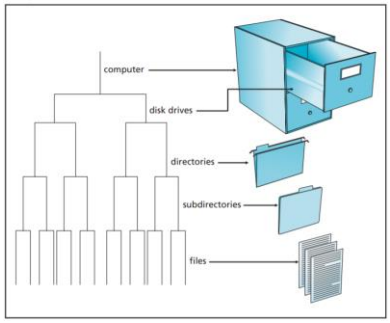
Different Types Data For You



How do you manage information?

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(Logical) File System



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(Physical) File System

Figure 10-3, Sectors are grouped into clusters on a hard disk

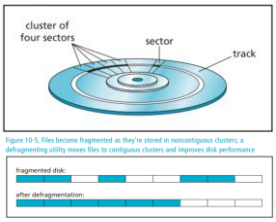
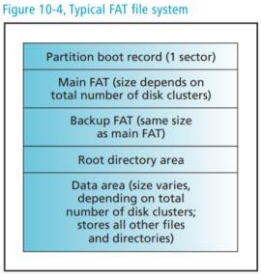


Figure 10-5, Files become fragmented as they're stored in noncontiguous clusters; a defragmenting utility moves files to contiguous clusters and improves disk performance

Figure 10-4, Typical FAT file system



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Homework

1. **Bitmap vs. Vector Images**
2. **(Office) Word vs. PDF File Formats**
3. **FAT vs. NTFS File System**
4. **Local vs. Cloud Storage**

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